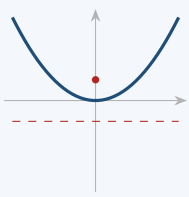
Parabolas: Focus & Directrix

Conic Sections — one parabola, four directions

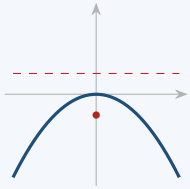
Answer Key with complete solutions

The big idea

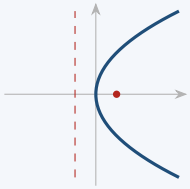
A **parabola** is every point that is the **same distance** from a point (the **focus**) and a line (the **directrix**). The **vertex** is halfway between them, and p is the distance (with a sign) from the vertex to the focus.



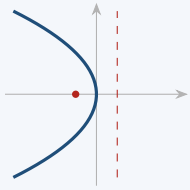
opens up



opens down



opens right



opens left

	Equation	Focus	Directrix	Axis
Up ($p > 0$) / Down ($p < 0$)	$(x-h)^2 = 4p(y-k)$	$(h,\ k+p)$	$y = k-p$	$x = h$
Right ($p > 0$) / Left ($p < 0$)	$(y-k)^2 = 4p(x-h)$	$(h+p,\ k)$	$x = h-p$	$y = k$

Simple case first. $x^2 = 4y$: here $4p = 4$, so $p = 1$. Vertex $(0,0)$, focus $(0,1)$, directrix $y = -1$. Check with the point $(2,1)$ on the curve: distance to the focus is 2, distance to the line $y = -1$ is 2. ✓

Watch out: The squared letter tells the direction. $x^2 \Rightarrow$ opens up or down; $y^2 \Rightarrow$ opens right or left. Then the sign of p decides which way. And $4p$ is not p: in $x^2 = 12y$, $p = 3$, not 12.

0 Warm-Up: Algebra 2 skills you'll need

MEDIUM

These come from earlier Algebra 2 units. Each one shows up again below.

1. **Distance formula.** Find the distance between $(-2,5)$ and $(4,-3)$.

   $\sqrt{(4-(-2))^2 + (-3-5)^2} = \sqrt{36+64} = \sqrt{100} = \mathbf{10}$.

2. **Distance formula.** The point $(x,4)$ is 5 units from $(1,0)$. Find every possible x.

   $(x-1)^2 + (4-0)^2 = 25 \Rightarrow (x-1)^2 = 9 \Rightarrow x - 1 = \pm 3 \Rightarrow \mathbf{x = 4}$ or $\mathbf{x = -2}$.

3. **Completing the square.** Write $y = x^2 + 6x + 5$ in vertex form and give the vertex.

   $y = (x^2 + 6x + 9) + 5 - 9 = \mathbf{(x+3)^2 - 4}$. Vertex $(\mathbf{-3}, \mathbf{-4})$.

4. **Completing the square.** Write $y = 2x^2 - 8x + 5$ in vertex form.

$y = 2(x^2 - 4x) + 5 = 2(x^2 - 4x + 4) + 5 - 8 = \mathbf{2(x-2)^2 - 3}$. Vertex $(2, -3)$.

5. **Transformations.** Describe $g(x) = -\frac{1}{2}(x+1)^2 + 4$ compared with $y = x^2$, and give its vertex.

Left 1, up 4, reflected over the x-axis (opens **down**), vertically compressed by $\frac{1}{2}$. Vertex $(\mathbf{-1, 4})$.

6. **Solving quadratics.** Solve $2x^2 - 3x - 5 = 0$.

$(2x-5)(x+1) = 0 \Rightarrow \mathbf{x = \frac{5}{2}}$ or $\mathbf{x = -1}$.

7. **Linear–quadratic system.** Solve $y = x^2 - 2$ and $y = x + 4$.

$x^2 - 2 = x + 4 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x-3)(x+2) = 0$. $x = 3$: $y = 7$; $x = -2$: $y = 2$. $(\mathbf{3, 7})$ and $(\mathbf{-2, 2})$.

8. **Inverse functions.** $f(x) = (x-1)^2 + 2$ for $x \geq 1$. Find $f^{-1}(x)$ and its domain. (Its graph is half of a sideways parabola.)

Swap: $x = (y-1)^2 + 2 \Rightarrow (y-1)^2 = x - 2 \Rightarrow y - 1 = \sqrt{x-2}$ (positive root, since $y \geq 1$).
$\mathbf{f^{-1}(x) = 1 + \sqrt{x-2}}$, domain $\mathbf{x \geq 2}$.

A Opens Up $(x-h)^2 = 4p(y-k),\ p > 0$

Example · opens up

Find the vertex, focus and directrix of $(x-1)^2 = 4(y+2)$.
Vertex $(h, k) = (1, -2)$. $4p = 4$, so $p = 1$ (positive: opens up). Focus $(1, -2+1) = (1, -1)$.
Directrix $y = -2 - 1 = -3$.

9. EASY Find the vertex, focus and directrix of $x^2 = 8y$.

$4p = 8 \Rightarrow p = 2$. Vertex $(\mathbf{0, 0})$, focus $(\mathbf{0, 2})$, directrix $\mathbf{y = -2}$.

10. EASY Find the focus and directrix of $x^2 = 12y$.

$4p = 12 \Rightarrow p = 3$. Focus $(\mathbf{0, 3})$, directrix $\mathbf{y = -3}$.

11. EASY Write the equation of the parabola with vertex $(0, 0)$ and focus $(0, 5)$.

Focus above the vertex: opens up, $p = 5$. $x^2 = 4(5)y$: $\mathbf{x^2 = 20y}$.

12. MEDIUM Find the vertex, focus, directrix and axis of symmetry of $(x-2)^2 = 8(y+1)$.

Vertex $(\mathbf{2, -1})$. $4p = 8 \Rightarrow p = 2$. Focus $(2, -1+2) = (\mathbf{2, 1})$. Directrix $y = -1 - 2$: $\mathbf{y = -3}$.
Axis $\mathbf{x = 2}$.

13. HARD Write $y = \frac{1}{4}x^2 - x + 3$ in the form $(x-h)^2 = 4p(y-k)$. Then find the vertex, focus and directrix.

Multiply by 4: $4y = x^2 - 4x + 12 \Rightarrow x^2 - 4x = 4y - 12$.
Complete the square: $x^2 - 4x + 4 = 4y - 12 + 4 \Rightarrow \mathbf{(x-2)^2 = 4(y-2)}$.
Vertex $(\mathbf{2,2})$, $p = 1$, focus $(\mathbf{2,3})$, directrix $\mathbf{y = 1}$. (Check: $x = 2$ gives $y = 1 - 2 + 3 = 2$ ✓)

B Opens Down $(x-h)^2 = 4p(y-k),\ p < 0$

Example · opens down

Find the vertex, focus and directrix of $x^2 = -8y$.
$4p = -8$, so $p = -2$ (negative: opens down). Vertex $(0,0)$, focus $(0,-2)$, directrix $y = 2$ — the directrix is on the *opposite* side.

14. EASY Find the vertex, focus and directrix of $x^2 = -16y$.

$4p = -16 \Rightarrow p = -4$. Vertex $(\mathbf{0,0})$, focus $(\mathbf{0,-4})$, directrix $\mathbf{y = 4}$.

15. EASY Find the focus and directrix of $x^2 = -4y$.

$4p = -4 \Rightarrow p = -1$. Focus $(\mathbf{0,-1})$, directrix $\mathbf{y = 1}$.

16. EASY Write the equation of the parabola with vertex $(0,0)$ and directrix $y = 3$.

The directrix is above, so the parabola opens down: $p = -3$. $x^2 = 4(-3)y$: $\mathbf{x^2 = -12y}$.

17. MEDIUM Write the equation of the parabola with focus $(1,-2)$ and directrix $y = 4$.

The vertex is halfway between: $(1, \frac{-2+4}{2}) = (1,1)$. The focus is 3 below the vertex: $p = -3$.
$\mathbf{(x-1)^2 = -12(y-1)}$.

18. HARD Write $x^2 + 6x + 8y + 1 = 0$ in standard form. Find the vertex, focus and directrix.

$x^2 + 6x = -8y - 1 \Rightarrow x^2 + 6x + 9 = -8y - 1 + 9 = -8y + 8 \Rightarrow \mathbf{(x+3)^2 = -8(y-1)}$.
Vertex $(\mathbf{-3,1})$, $4p = -8 \Rightarrow p = -2$, focus $(\mathbf{-3,-1})$, directrix $\mathbf{y = 3}$.

C Opens Right $(y-k)^2 = 4p(x-h),\ p > 0$

Example · opens right

Find the vertex, focus and directrix of $y^2 = 12x$.
y is squared: it opens sideways. $4p = 12$, so $p = 3$ (positive: right). Vertex $(0,0)$, focus $(3,0)$, directrix $x = -3$ (a *vertical* line).

19. EASY Find the vertex, focus and directrix of $y^2 = 4x$.

$4p = 4 \Rightarrow p = 1$. Vertex $(\mathbf{0,0})$, focus $(\mathbf{1,0})$, directrix $\mathbf{x = -1}$.

20. EASY Find the focus and directrix of $y^2 = 20x$.

$4p = 20 \Rightarrow p = 5$. Focus $(\mathbf{5}, \mathbf{0})$, directrix $\mathbf{x = -5}$.

21. EASY Write the equation of the parabola with vertex $(0, 0)$ and focus $(2, 0)$.

Focus to the right: y^2 form, $p = 2$. $\mathbf{y^2 = 8x}$.

22. MEDIUM Find the vertex, focus, directrix and axis of symmetry of $(y - 3)^2 = 12(x + 2)$.

Vertex $(\mathbf{-2}, \mathbf{3})$. $4p = 12 \Rightarrow p = 3$. Focus $(-2 + 3,\ 3) = (\mathbf{1}, \mathbf{3})$. Directrix $x = -2 - 3$: $\mathbf{x = -5}$. Axis $\mathbf{y = 3}$.

23. HARD **Satellite dish.** A dish's cross-section is a parabola with vertex at the origin, opening right. The dish is 24 inches wide and 4 inches deep. How far from the vertex should the receiver (at the focus) be placed?

The rim is at $x = 4$ (depth), $y = \pm 12$ (half the width), so $(4, 12)$ is on $y^2 = 4px$.
$144 = 4p(4) = 16p \Rightarrow p = 9$. The focus is $(9, 0)$: the receiver goes **9 inches** from the vertex.

D Opens Left $(y - k)^2 = 4p(x - h),\ p < 0$

Example · opens left

Find the vertex, focus and directrix of $(y + 1)^2 = -4(x - 2)$.
Vertex $(2, -1)$. $4p = -4$, so $p = -1$ (negative: left). Focus $(2 - 1,\ -1) = (1, -1)$. Directrix $x = 2 + 1 = 3$.

24. EASY Find the vertex, focus and directrix of $y^2 = -8x$.

$4p = -8 \Rightarrow p = -2$. Vertex $(\mathbf{0}, \mathbf{0})$, focus $(\mathbf{-2}, \mathbf{0})$, directrix $\mathbf{x = 2}$.

25. EASY Find the focus and directrix of $y^2 = -24x$.

$4p = -24 \Rightarrow p = -6$. Focus $(\mathbf{-6}, \mathbf{0})$, directrix $\mathbf{x = 6}$.

26. EASY Write the equation of the parabola with vertex $(0, 0)$ and directrix $x = 7$.

The directrix is on the right, so it opens left: $p = -7$. $\mathbf{y^2 = -28x}$.

27. MEDIUM Write the equation of the parabola with vertex $(3, -1)$ and focus $(1, -1)$. Give its directrix.

The focus is 2 to the left: opens left, $p = -2$. $\mathbf{(y + 1)^2 = -8(x - 3)}$. Directrix $x = 3 + 2$: $\mathbf{x = 5}$.

28. HARD Write $y^2 - 4y + 4x + 12 = 0$ in standard form. Find the vertex, focus and directrix, then check one point with the definition.

$y^2 - 4y = -4x - 12 \Rightarrow y^2 - 4y + 4 = -4x - 8 \Rightarrow \mathbf{(y - 2)^2 = -4(x + 2)}$.
Vertex $(\mathbf{-2}, \mathbf{2})$, $p = -1$, focus $(\mathbf{-3}, \mathbf{2})$, directrix $\mathbf{x = -1}$.
Check: $x = -3$ gives $(y - 2)^2 = 4$, so $(-3, 4)$ is on it. Distance to the focus: 2; to the line $x = -1$:

2. ✓